## Lesson 3

# Dimensionality Reduction & Latent Variables

> *"It can scarcely be denied that the supreme goal of all theory is to make the irreducible basic elements as simple and as few as possible without having to surrender the adequate representation of a single datum of experience."*
>
> — Albert Einstein, *On the Method of Theoretical Physics* (1933)

## Introduction & Motivation

The first two lessons built the statistical foundation for supervised learning: expected error, empirical risk, regression, classification, optimization, and validation. We now encounter a very specific engineering failure mode: the data matrix may contain many sensor columns, but the process itself may have only a few true physical degrees of freedom.

This lesson introduces the language of **latent variables**. Instead of treating every measured signal as independent, we learn how to compress correlated measurements into a smaller number of orthogonal coordinates that preserve the dominant physical information. This is the central idea behind chemometrics, process monitoring, and modern digital twins.

**Lesson Objectives:**

- Understand why multi-collinearity makes the OLS inverse unstable in high-density sensor arrays.
- Derive Principal Component Analysis (PCA) from a constrained variance-maximization problem.
- Interpret eigenvectors, eigenvalues, and scree plots as physical latent-state diagnostics.
- Explain why PCA can fail for prediction and why Partial Least Squares (PLS) is often preferred in chemical engineering.
- Connect dimensionality reduction to a realistic electrolyzer sensor-

fusion problem.

**The Physical Reality: Sensors Are Not Independent Physics**

A plant historian may store hundreds of tags, but the reactor is still governed by a small number of balances, constraints, and state variables. Dimensionality reduction is not only a computational trick; it is an attempt to recover the hidden low-dimensional physics buried under redundant instrumentation.

## 3.1 The Curse of Dimensionality & Multi-Collinearity

In Lesson 2, we derived the Ordinary Least Squares (OLS) analytical solution for linear regression: $\hat{\beta} = (X^T X)^{-1} X^T Y$. While mathematically sound, this equation harbors a fatal vulnerability when applied to real-world process engineering data.

Modern chemical plants stream data from high-density sensor arrays (e.g., $p = 200$ sensors). However, the true physical degrees of freedom are strictly limited. According to Gibbs' Phase Rule and the fundamental conservation laws of mass, momentum, and energy, the actual number of independent thermodynamic states governing the process is very small (often $k \leq 5$).

Because the sensors are measuring the exact same underlying physical phenomena, their readings are highly correlated. This is known as **Multi-Collinearity**.

When columns in our design matrix $X \in \mathbb{R}^{N \times p}$ are linearly dependent, the matrix $(X^T X)$ becomes *singular* or ill-conditioned. Its determinant approaches zero, meaning the inverse $(X^T X)^{-1}$ mathematically explodes to infinity. Consequently, the estimated parameters $\hat{\beta}$ become extraordinarily sensitive to microscopic sensor noise, resulting in violent, physically impossible predictions.

To build robust models, we must compress the high-dimensional, noisy sensor space $\mathcal{X}$ into a low-dimensional, orthogonal **Latent Space** $\mathcal{Z}$.

## 3.2 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is the foundational algorithm for unsupervised dimensionality reduction. Its objective is to find a set of

orthogonal axes (directions) in the input space along which the variance of the data is maximized.

Let $X \in \mathbb{R}^{N \times p}$ be our mean-centered data matrix (where the empirical mean of each sensor column has been subtracted so that $\mu = 0$). The empirical covariance matrix is defined as:

$$S = \frac{1}{N-1} X^T X \tag{3.1}$$

where $S \in \mathbb{R}^{p \times p}$ is a symmetric, positive semi-definite matrix.

We seek a unit projection vector $v \in \mathbb{R}^p$ (such that $\|v\|_2 = 1$) that maximizes the variance of the projected data $Z = Xv$. The variance of this projection is given by:

$$\text{Var}(Xv) = \frac{1}{N-1}(Xv)^T(Xv) = v^T S v \tag{3.2}$$

To maximize Equation 3.2 subject to the unit-length constraint $v^T v = 1$, we formulate the constrained optimization problem using a **Lagrangian**:

$$\mathcal{L}(v, \lambda) = v^T S v - \lambda(v^T v - 1) \tag{3.3}$$

where $\lambda \in \mathbb{R}$ is the Lagrange multiplier. Taking the gradient with respect to the vector $v$ and setting it to the zero vector yields:

$$\begin{aligned} \nabla_v \mathcal{L}(v, \lambda) &= 2Sv - 2\lambda v = 0 \\ Sv &= \lambda v \end{aligned} \tag{3.4}$$

Equation 3.4 is the strict mathematical definition of **Eigenvalue Decomposition**. We have rigorously proven that the direction $v$ maximizing the variance is precisely the *eigenvector* of the covariance matrix $S$. The corresponding *eigenvalue* $\lambda$ represents the exact amount of variance captured along that component.

By sorting the eigenvectors $\{v_1, v_2, \ldots, v_p\}$ in descending order of their eigenvalues $\{\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p\}$, we define the **Principal Components**.

**Definition: Latent Space Projection**

To reduce the system from $p$ correlated sensors to $k$ independent latent variables ($k \ll p$), we construct a projection matrix $V_k \in \mathbb{R}^{p \times k}$ containing the first $k$ eigenvectors. The transformed latent state matrix $Z \in \mathbb{R}^{N \times k}$ is:

$$Z = XV_k \tag{3.5}$$

Because the columns of $Z$ are perfectly orthogonal, we have completely eliminated multi-collinearity.

### 3.2.1 Reconstruction Error and Choosing the Number of Components

PCA is useful only if we choose $k$ intelligently. If $k$ is too small, we discard real physical structure. If $k$ is too large, we keep redundant noise and reintroduce the high-variance behavior we were trying to avoid.

Given the retained projection matrix $V_k$, the rank-$k$ reconstruction of the original centered data is:

$$\hat{X}_k = ZV_k^T = XV_kV_k^T \tag{3.6}$$

The reconstruction error is therefore:

$$\|X - \hat{X}_k\|_F^2 \tag{3.7}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. A small reconstruction error means the latent variables preserve most of the measured process behavior.

The fraction of variance explained by the first $k$ components is:

$$\text{Explained Variance}(k) = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{p} \lambda_j} \tag{3.8}$$

In engineering practice, we often select $k$ at the "elbow" of the scree plot: the point after which adding another component mostly captures sensor noise rather than a meaningful physical degree of freedom.

## 3.3 Partial Least Squares (PLS) Regression

While PCA is exceptionally powerful for identifying structure, it has a fatal flaw when used as a precursor to regression.

**Q&A: The Flaw in PCA for Regression**
**Question:** PCA strictly maximizes the variance within the input matrix $X$. Why might this cause our predictive regression model to fail?
**Answer:** PCA is completely blind to the target variable $Y$. The direction of maximum variance in $X$ might be dominated by irrelevant background noise (e.g., ambient room temperature fluctuating). Meanwhile, the chemical reaction yield ($Y$) might depend on a subtle trace variable with very low variance. PCA will discard the critical trace variable, destroying the model's predictive power.

To solve this, chemical engineers and chemometricians utilize **Partial Least Squares (PLS) Regression**. Unlike PCA, which only maximizes the variance in $X$ ($v^T X^T X v$), PLS seeks projection vectors $w \in \mathbb{R}^p$ for $X$ and $c \in \mathbb{R}$ for $Y$ that *maximize the covariance* between the projected inputs and outputs. Mathematically, the first PLS component solves:

$$\underset{\|w\|=1, \|c\|=1}{\text{argmax}} \quad \text{Cov}(Xw, Yc) \tag{3.9}$$

By maximizing covariance, PLS ensures that the newly created latent variables are orthogonal and engineered to be highly predictive of $Y$.

## 3.4 Case Study: Sensor Fusion in Electrolyzers

**Case Study: PEM Electrolyzer Latent States**

**The Setup:** A process engineer is monitoring the health of a 5-Megawatt Proton-Exchange Membrane (PEM) electrolyzer stack producing green hydrogen. The stack is equipped with $p = 100$ sensors (voltage taps and thermocouples across every cell).
**Cross-Disciplinary Reading:** This is a high-density sensor-array problem. The same issue appears in battery packs, turbine monitoring, structural health sensing, semiconductor tools, and robotics: many sensors report highly correlated information because they observe the same underlying physical state.
**The Collinearity Problem:** Because the cells operate in series, the 100 voltage sensors move simultaneously with load changes. Attempting

to run a standard regression model to predict stack efficiency ($Y$) using all 100 raw sensors ($X$) fails because the matrix $(X^T X)$ is nearly singular.
**The ML Solution:** The engineer applies PCA to the $N \times 100$ data matrix. They construct a *Scree Plot* and discover that just two eigenvalues ($\lambda_1$ and $\lambda_2$) account for 98.5% of the total system variance.

- **PC1:** Mathematically represents the primary variance axis. Physically, it correlates perfectly with the overall current load and global membrane hydration.
- **PC2:** Mathematically captures the orthogonal secondary variance. Physically, it isolates the subtle differential temperature gradients caused by gas crossover.

By compressing $X$ into $Z \in \mathbb{R}^{N \times 2}$, the engineer builds a perfectly stable, physically interpretable Digital Twin.

As shown in Figure 3.1, Panel A, the raw sensors are highly correlated. Rather than feeding redundant inputs into a model, PCA identifies the orthogonal axes of physical variance. Panel B (the Scree Plot) proves that just 2 latent variables capture 98.5% of the system's thermodynamic information, allowing us to safely discard dimensions 3 through 100 as irreducible noise.

## 3.5 Summary

In this lesson, we established the mathematical tools to rescue physical models from the curse of high-dimensional sensor data:

- **The Multi-Collinearity Problem:** We proved that highly correlated engineering sensors cause the OLS regression matrix $(X^T X)$ to become singular, resulting in catastrophic model instability.
- **Mathematical Foundations of PCA:** By formulating a constrained Lagrangian optimization (Equation 3.3), we proved that the eigenvectors of the covariance matrix (Equation 3.4) form the optimal orthogonal axes to maximize data variance, and used reconstruction error (Equation 3.7) plus explained variance (Equation 3.8) to choose $k$.

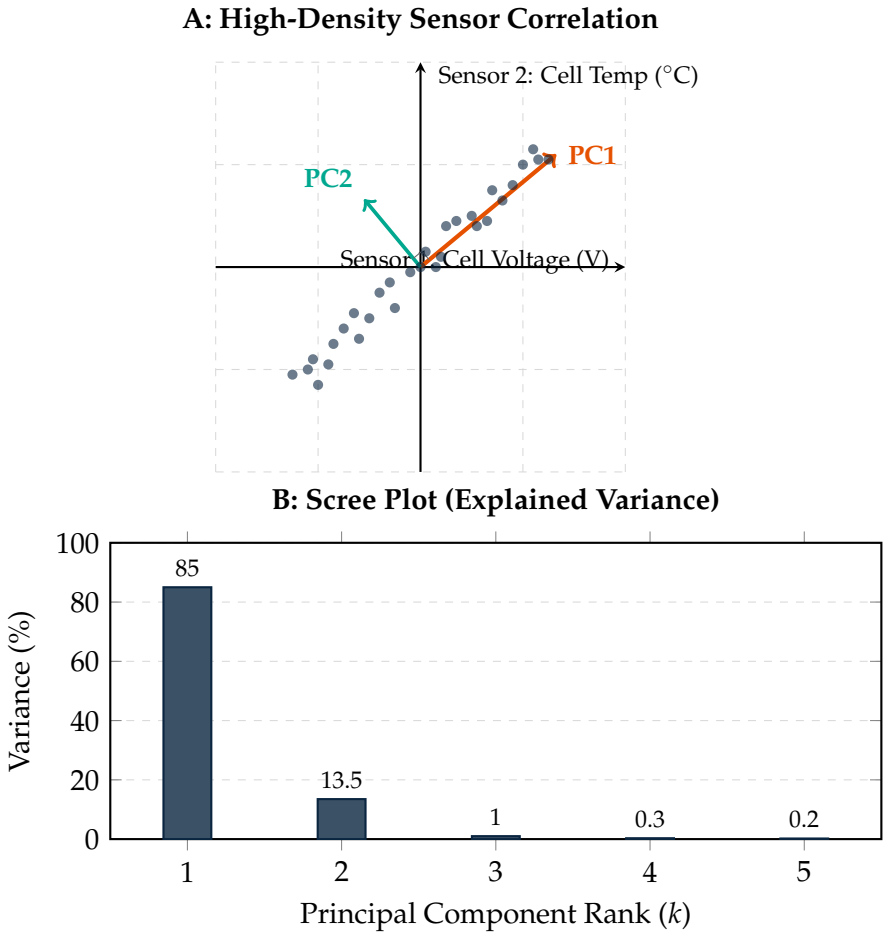



Figure 3.1: Principal Component Analysis of a high-density sensor array. Panel A shows that the raw voltage and temperature sensors are strongly correlated, while PCA rotates the coordinate system into two exactly orthogonal principal axes, PC1 and PC2. Panel B shows the explained-variance scree plot, demonstrating that the first two principal components capture 98.5% of the system variance.

- **The Necessity of PLS:** We highlighted the fatal flaw of PCA for regression (its blindness to the target $Y$), and introduced Partial Least Squares (Equation 3.9) as the chemometric standard that maximizes covariance between inputs and outputs.
- **Engineering Application:** Through the PEM Electrolyzer case study and the Scree Plot visualization, we demonstrated that mathematical Principal Components correspond directly to hidden physical states (Latent Variables) governed by thermodynamic laws.